Dr. Wai Hnin Eaindrar Mg

Chief AI Engineer & Computer Vision Research Scientist (Ph.D. in Engineering) | Multimodal AI (VLM / VLA / LLM) & Edge AI

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EXECUTIVE SUMMARY & CORE COMPETENCIES

Results-driven **AI Research Scientist and Chief Computer Vision Engineer (Ph.D. in Engineering)** with **5+ years of end-to-end R&D and commercial production experience** spanning state-of-the-art **Multimodal Foundation Models (VLMs / VLAs / LLMs)**, **Real-Time Multi-Camera Multi-Object Tracking**, **3D/2D Pose & Action Recognition**, and **Ultra-Low-Latency Edge AI Deployment (NVIDIA Jetson / TensorRT / ONNX)**. Inventor of the patented **Customized Tracking Algorithm (CTA)** (achieving **100% ID tracking accuracy** under severe multi-camera occlusion), first author of **6 peer-reviewed journal papers (Impact Factor up to 8.9)**, and recipient of the **ICGEC 2024 Best Paper Presentation Award**. Proven track record of bridging mathematical algorithm formulation with full-stack software engineering (Python, C/C++, FastAPI, PyQt/Tkinter GUI, Alert APIs) and pre-sales client consulting to deliver mission-critical AI products.

- **Multimodal AI & Action Understanding:** PyTorch fine-tuning (LoRA / PEFT) of Transformer-based VLMs, VLAs, and LLMs with Prompt Engineering and RAG for zero/few-shot object reasoning, retail anomaly detection (shoplifting, fighting, item picking), and industrial packing sequence recognition.
- **Multi-Camera Vision & Occlusion-Robust Tracking (CTA):** Object detection & instance segmentation (YOLO, CNNs), masked/eyeglass-resilient face recognition, emotion/gender/age estimation, Human Re-ID, multi-camera geometric calibration, and zero-annotation keypoint fall detection up to ~25m.
- **Edge AI Acceleration & Production Systems:** Model quantization, pruning, and inference acceleration via CUDA, TensorRT, and ONNX on NVIDIA Jetson Nano / Orin Nano; multi-threaded parallel camera pipelines, custom GUI applications, and real-time hardware alert APIs (Patlite, LineEyes, HTTP, Email).

WORK EXPERIENCE

AIRUCA Inc. (AI Software & Edge AI Hardware R&D — Capital: ¥269.15M, 15 Employees) **Apr. 2025 – Present**

Chief Engineer (AI R&D Lead) & Principal AI Researcher *Tokyo, Japan*

- **Project 1 — Multi-Camera People Analytics, Masked/Eyeglass Face Recognition & CTA Multi-Tracking System:**
  - Architected a commercial real-time multi-camera People Analytics platform capable of accurate facial identification even when subjects wear **surgical masks, eyeglasses, or sunglasses**, integrated with real-time **emotion recognition** and **gender/age classification (>90% accuracy)**.
  - Engineered the proprietary **Customized Tracking Algorithm (CTA)** to dynamically bind local Track-IDs with global Face-IDs across overlapping and side-view camera arrays using custom multi-threading and frame-level synchronization, completely eliminating ID switching during heavy crowd occlusion.
  - Developed a full-featured production GUI (automated multi-camera registration, unknown-visitor enrollment, live recognition feed, person search) and multi-channel alert APIs (**Patlite revolving lights, LineEyes, HTTP webhooks, Email**).
  - **Key Results:** Achieved **100% face recognition & multi-person tracking accuracy** on primary multi-camera feeds (**98% on steep side-view cameras**) and **>95% behavior classification accuracy**; deployed as a flagship commercial product.
- **Project 2 — Long-Range (~25m) AI Fall Detection System (Version 1 & Version 2):**
  - Designed **Version 1** utilizing high-speed YOLO bounding-box aspect-ratio transitions and motion-vector kinematics for lightweight real-time fall detection.
  - Spearheaded **Version 2** by integrating **intrinsic/extrinsic camera calibration with 2D/3D skeletal keypoints detection**, formulating a geometric posture-collapse algorithm that requires **zero manual fall dataset annotations** and generalizes directly to unseen camera perspectives.
  - **Key Results:** Solved small-target degradation at distances up to **~25 meters** in wide-angle CCTV feeds; integrated real-time emergency dispatch across HTTP, Patlite, LINE EYES, voice alarms, and email.
- **Project 3 — Retail Shopping Basket, Item & Fine-Grained Sticker Recognition with Continuous Tracking:**
  - Built a multi-stage detection, fine-grained classification, and continuous tracking pipeline for **shopping baskets, diverse retail merchandise items, and small-ROI discount/identification stickers**, augmented by VLM-LLM visual-textual feature alignment for context-aware product verification.
  - **Key Results:** Delivered robust simultaneous tracking of baskets, items, and sub-centimeter discount stickers under rapid movement and hand occlusion, automating retail checkout auditing.
- **Project 4 — Real-Time Packing Action Recognition & Multimodal Behavior Recognition (VLM / VLA / Edge AI):**
  - Engineered a **Real-Time Packing Action Recognition** engine classifying sequential industrial packaging operations (***cutting, opening, closing***) and a **Human Behavior Recognition** system fusing **Face-ID + Full-Body Re-ID** to log worker/customer actions and detect anomalies (shoplifting, item picking, book reading, fighting).
  - Fine-tuned transformer-based **Vision-Language Models (VLMs)** and **Vision-Language-Action (VLA)** architectures in PyTorch using **LoRA / parameter-efficient fine-tuning**, multimodal feature fusion, prompt engineering, and **Retrieval-Augmented Generation (RAG)**; accelerated real-time edge inference on **NVIDIA Jetson Nano & Jetson Orin Nano** via CUDA, TensorRT, and ONNX.
- **Project 5 — Pre-Sales Technical Consulting, PoC Architecture & On-Site Field Deployment:**
  - Collaborated directly with the sales team as the lead technical authority in customer discovery meetings—conducting technical feasibility assessments, designing PoC architectures, performing **on-site software installation & camera calibration**, and executing rapid system/code customizations.
  - **Stack:** Python; C/C++; PyTorch; OpenCV; YOLO; Keypoints Detection; Camera Calibration; VLMs / VLAs / LLMs; LoRA; RAG; NVIDIA Jetson (Nano / Orin Nano); CUDA; TensorRT; ONNX; FastAPI; PyQt/Tkinter; Multi-threading; Linux; Windows; Git.

WORK EXPERIENCE (CONTINUED)

University of Miyazaki — Visual Information Laboratory (Thi Thi LAB) **Apr. 2022 – Mar. 2025**
Doctoral Researcher & Lead Algorithm Developer (Ph.D. Candidate in Engineering) *Miyazaki, Japan*

- Architected an end-to-end, 24/7 real-time **4-camera smart livestock monitoring, behavior analysis, and calving-time prediction system** to automate dairy/beef farm management and prevent calf mortality from dystocia (difficult birth).
- **High-Precision Detection & Segmentation:** Designed custom deep learning (YOLO/CNN) face, ear-tag, and full-body detection & instance segmentation frameworks that filter out complex farm background noise (workers, tractors, lighting shifts), achieving **99.81% detection & segmentation accuracy** across continuous day and night infrared/RGB video streams.
- **Invention of Customized Tracking Algorithm (CTA) & Global Re-ID:** Formulated and implemented **CTA** to resolve severe inter-animal occlusions and posture deformations by coupling local trajectory Track-IDs with deep-learning/machine-learning Global Identification IDs, achieving **99.11% tracking accuracy** and **99.18% global re-identification accuracy** across twenty 12-hour continuous farm trials.
- **Multi-Modal Time-Series & Signal-Based Calving Prediction (4 Frameworks + Spectral Analysis):** Engineered four complementary predictive modeling paradigms to forecast exact calving time and classify normal vs. abnormal (human-assistance required) deliveries:
  - **Approach 1 (AI Movement Rate Variability — MRV):** Quantified fine-grained trajectory velocity fluctuations around parturition, predicting calving **3 hours in advance with 100% accuracy**.
  - **Approach 2 (Behavioral State & Posture Classification):** Classified standing/lying transitions, lameness, and mounting behaviors in real time to predict calving status **3 hours in advance**.
  - **Approach 3 (Statistical Trajectory Feature Modeling):** Formulated statistical movement descriptors to classify cattle into normal vs. dystocia/assistance-needed groups and predict calving **at least 6 hours in advance**.
  - **Approach 4 (Custom LSTM with Euclidean Fluctuating Summation [EFS] & SIFT/CWT Spectral Analysis):** Designed an EFS-embedded LSTM time-series forecaster resilient to sensor dropouts (predicting **4 hours in advance**) and pioneered **SIFT Spectrogram & Continuous Wavelet Transform (CWT) Scalogram** trajectory signal analysis.
  - **Key Results & Stack:** Filed **1 Japanese Patent (2025-007606)**; published **6 first-author journal papers (up to IF: 8.9)**; won **ICGEC 2024 Best Paper Presentation Award**. | **Stack:** Python; PyTorch; OpenCV; YOLO; Custom LSTM (EFS); SIFT/CWT Signal Processing; Multi-Camera Calibration; MATLAB; Linux; Git.

Frontiir Myanmar Co. Ltd (Leading ISP & IT Infrastructure Provider — ~2,000 Employees) **Nov. 2021 – Mar. 2022**
Senior Associate Engineer – Data Analyst *Yangon, Myanmar*

- **Customer Behavior & Churn Predictive Modeling:** Engineered machine learning classification and predictive models directly from terabyte-scale raw binary network telemetry to segment subscribers into *Loyal, Good, and At-Risk/Churn* tiers, achieving **92% classification accuracy**.
- **Data Pipeline Architecture & BI Leadership:** Re-architected automated ETL workflows using **Apache Airflow**, cutting end-to-end data processing latency by **55%**; designed interactive executive BI dashboards that drove a **20% increase in customer retention**, and mentored junior data analysts.
  - **Stack:** Python; Apache Airflow; Scikit-learn; SQL; Statistical Modeling; Tableau / PowerBI; Linux; Windows.

Freelance AI Consultant & Computer Vision Engineer **Dec. 2020 – Oct. 2021**
Independent Computer Vision Architect *Yangon, Myanmar*

- **Multi-Person Dance Motion & Pose Tracking:** Designed a real-time multi-person dancing detection, skeletal tracking, and performance scoring system achieving **98% individual dancer tracking accuracy** under rapid motion and crossover.
- **Large-Scale Surveillance Face Recognition & Cross-Camera Human Re-ID:** Built a multi-face detection, tracking, and recognition engine (**97% accuracy** across diverse indoor/outdoor lighting) and a deep **Human Re-Identification (Re-ID)** model across non-overlapping CCTV camera networks, improving client system reliability by **80%**. | **Stack:** Python; PyTorch; OpenCV; YOLO; CNNs; Human Re-ID; Linux.

Metro Computer Myanmar Co. Ltd (Software & IT Solutions — ~50 Employees) **Sep. 2019 – Nov. 2020**
AI Engineer (Computer Vision & GUI Systems) *Yangon, Myanmar*

- **Real-Time Security Face Recognition & Automated Attendance GUI:** Designed single- and multi-face recognition desktop applications with custom **PyQt / Tkinter GUIs** (**95% identification accuracy** using parallel processing) and automated staff attendance logging, **reducing manual entry errors by 90%**.
- **Retail Shoplifting Detection & Restaurant Occupancy Analytics:** Engineered real-time suspicious-behavior/theft recognition (**reducing potential store theft losses by 85%**) and automated restaurant table-counting software (**cutting customer wait times by 80%**). | **Stack:** Python; C++; OpenCV; Deep Learning; PyQt; Tkinter; Linux; Windows.

EDUCATION

University of Miyazaki **Apr. 2022 – Mar. 2025**
Doctor of Philosophy (Ph.D.) in Engineering — Computer Science and Bioinformatics *Miyazaki, Japan*

- **Dissertation:** *A Study of Real-time Cattle Monitoring System for Calving Time Prediction* | **Lab:** Visual Information Laboratory (Thi Thi LAB).

University of Technology (Yatanarpon Cyber City — UTYCC) **May 2017 – Feb. 2020**
Master of Engineering (M.Eng.) — Information Science and Technology (IST) *Pyin Oo Lwin, Myanmar*

- **Thesis:** *Face Sketch Construction System Using Graphics Pipeline Algorithm.*

University of Technology (Yatanarpon Cyber City — UTYCC) **Nov. 2011 – Feb. 2017**
Bachelor of Engineering (B.Eng.) — Information Science and Technology (IST) *Pyin Oo Lwin, Myanmar*

No. (6) Basic Education High School **Jun. 2006 – May 2011**
High School Diploma (Science Stream) — Matriculation Exam with 4 Distinctions (Mathematics, Physics, Chemistry, Biology) *Pathein, Myanmar*

PATENTS & PEER-REVIEWED PUBLICATIONS (6 JOURNALS, 5 CONFERENCES)

- **[Patent]** Thi Thi Zin, Pyke Tin, and **Wai Hnin Eaindrar Mg**, "Cattle Dystocia Prediction System (牛の難産予測システム)", **Japan Patent Application No. 2025-007606**, filed Jan. 20, 2025 (Applicant: National University Corporation University of Miyazaki).
- **[J1] Wai Hnin Eaindrar Mg**, Pyke Tin, M. Aikawa, Y. Horii, K. Honkawa, and Thi Thi Zin, "Signal-Based Feature Analysis of Behavioral Trajectories for Predicting Calving Time and Classifying Assistance Needs", *Elsevier: Computers and Electronics in Agriculture*, vol. 243, 111301, 2026 (14 pages). DOI: https://doi.org/10.1016/j.compag.2025.111301 **[Impact Factor: 8.9]**
- **[J2] Wai Hnin Eaindrar Mg**, Thi Thi Zin, Pyke Tin, M. Aikawa, Y. Horii, and K. Honkawa, "Automated Cattle Monitoring System for Calving Time Prediction Using Trajectory Data Embedded Time Series Analysis", *IEEE Open Journal of the Industrial Electronics Society*, vol. 6, pp. 216–234, 2025 (16 pages). DOI: https://doi.org/10.1106/OJIES.2025.3533663 **[Impact Factor: 5.2]**
- **[J3] Wai Hnin Eaindrar Mg**, Thi Thi Zin, Pyke Tin, M. Aikawa, Y. Horii, and K. Honkawa, "Automated system for calving time prediction and cattle classification utilizing trajectory data and movement features", *Scientific Reports (Nature Portfolio)*, vol. 15, article no. 2378, 2025 (24 pages). DOI: https://doi.org/10.1038/s41568-025-85632-0 **[Impact Factor: 3.8]**
- **[J4] Wai Hnin Eaindrar Mg**, Pyke Tin, M. Aikawa, I. Kobayashi, Y. Horii, K. Honkawa, and Thi Thi Zin, "Customized Tracking Algorithm for Robust Cattle Detection and Tracking in Occlusion Environments", *Sensors (Basel, Switzerland)*, vol. 24, issue 4, article no. 1181, 2024 (26 pages). DOI: https://doi.org/10.3360/s24041181 **[Impact Factor: 3.4]**
- **[J5] Wai Hnin Eaindrar Mg** and Thi Thi Zin, "Cattle face detection with ear tags using YOLOv5 model", *ICIC Express Letters, Part B: Applications*, vol. 14, issue 1, pp. 65–72, 2023 (8 pages). DOI: https://doi.org/10.24507/icicelb.14.01.65 **[Impact Factor: 1.0]**
- **[J6] Wai Hnin Eaindrar Mg** and Tin Myint Naing, "Face Sketch Image Category Classification Using Bag of Words (BoW) Model", *Scientific Journal of Innovative Research*, vol. 1, issue 1, pp. 135–142, 2016 (8 pages). DOI: https://doi.org/10.5281/zenodo.14665503 **[Impact Factor: 1.0]**
- **[P1] Wai Hnin Eaindrar Mg**, Thi Thi Zin, Pyke Tin, M. Aikawa, Y. Horii, and K. Honkawa, "Utilizing Behavioral Features for Predicting Calving Time", *The 16th International Conference on Genetic and Evolutionary Computing (ICGEC 2024)*, Lecture Notes in Electrical Engineering 1321, vol. 1, pp. 148–156, Miyazaki, Japan, Aug. 26–28, 2024 (12 pages). DOI: https://doi.org/10.1007/678-681-66-1531-5_15 **[Best Paper Presentation Award]**
- **[P2] Wai Hnin Eaindrar Mg**, Pyke Tin, M. Aikawa, I. Kobayashi, Y. Horii, K. Honkawa, and Thi Thi Zin, "AI Driven Movement Rate Variability Analysis Around the Time of Calving Events in Cattle", *The 15th International Conference on Genetic and Evolutionary Computing (ICGEC 2023)*, Lecture Notes in Electrical Engineering 1114, vol. 2, pp. 227–237, Kaohsiung, Taiwan, Oct. 6–8, 2023 (11 pages). DOI: https://doi.org/10.1007/678-681-66-6412-0_24
- **[P3] Wai Hnin Eaindrar Mg** and Tin Myint Naing, "Face Sketch Construction System Using Graphics Pipeline Algorithm", *The 10th International Conference on Science and Engineering (ICSE 2016)*, pp. 270–274, Yangon Technological University, Myanmar, Dec. 7–8, 2016 (5 pages). DOI: https://doi.org/10.5281/zenodo.14665534
- **[C1] Wai Hnin Eaindrar Mg** and Thi Thi Zin, "Cattle Face Recognition and Ear Tags Identification", *5th International Symposium on Information and Knowledge Management (ISIKM 2022)*, Onsite (Guangzhou, China / Kumamoto, Japan) and Online, Mar. 25–26, 2022.
- **[Poster] Wai Hnin Eaindrar Mg**, Thi Thi Zin, Pyke Tin, M. Aikawa, I. Kobayashi, Y. Horii, and K. Honkawa, "Real-time cattle detection and calving time prediction using deep learning", *16th NOKOH Student Seminar in English*, University of Miyazaki, Japan, Nov. 7, 2023.

AWARDS, TECHNICAL SKILLS, OPEN-SOURCE & INTERESTS

- **Honors & Competitive Awards:** Best Paper Presentation Award (*The 16th International Conference on Genetic and Evolutionary Computing — ICGEC 2024*, Miyazaki, Japan); **First Prize** (*Face Reading for Future Prediction AI Competition*); **First Prize** (*Myanmar Freight Trading System Development Competition*); **Third Prize** (*UTYCC E-Library System & YCC Library Management System*).
- **Multimodal AI & Deep Learning:** Vision-Language Models (VLMs); Vision-Language-Action Models (VLAs); Large Language Models (LLMs); Parameter-Efficient Fine-Tuning (LoRA / PEFT); Prompt Engineering; Retrieval-Augmented Generation (RAG); Multimodal Feature Fusion; Transformers; CNNs; Custom LSTM Time-Series Forecasting (EFS); SIFT Spectrogram & CWT Wavelet Signal Processing.
- **Computer Vision & Video Analytics:** Object Detection & Instance Segmentation (YOLOv5/v8/Custom); 2D/3D Pose & Skeletal Keypoints Detection; Multi-Camera Geometric Calibration; Multi-Object Tracking (Proprietary CTA); Masked/Eyeglass Face Recognition; Emotion, Gender & Age Estimation; Full-Body Human & Animal Re-Identification (Re-ID); Real-Time Action & Behavior Recognition.
- **Edge AI, Programming & Production Engineering:** Python; C/C++; MATLAB; Java (J2SE, J2EE); JavaScript; PHP; SQL; PyTorch; OpenCV; Scikit-learn; FastAPI; CUDA; TensorRT; ONNX; NVIDIA Edge AI Platforms (Jetson Nano / Jetson Orin Nano); Multi-threading & Parallel Camera Synchronization; Apache Airflow; PyQt / Tkinter GUI; Hardware & Notification APIs (Patlite, LineEyes, HTTP, Email); Docker; Linux; Windows; Git.
- **Selected Open-Source GitHub Projects:** Automated-System-for-Calving-Time-Prediction-and-Cattle-Classification (Multi-camera livestock tracking & calving prediction); Multi-Face-Recognition (Parallel multi-face identification & tracking); Face-Recognition (Real-time security GUI); Dancing_Detection_Tracking (Multi-person pose & motion tracking); UserRegistrationDemo.
- **Languages:** English (Professional Working & International Academic Publishing); Japanese (Conversational — Daily Business & Technical Communication in Japan); Myanmar (Native); Chinese (Basic).
- **Leadership, Soft Skills & Interests:** Technical Team Leadership; Pre-Sales Client Consulting & PoC Design; Junior Engineer Mentoring; Academic Peer-Review & Technical Documentation | **Interests:** Benchmarking New Open-Source Multimodal Models on GitHub; Badminton; Walking; Traveling.

KEY DIFFERENTIATORS & ENGINEERING VALUE PROPOSITION

- **1. Bridging State-of-the-Art Multimodal AI Research (VLM / VLA / LLM) with Industrial Deployment:** Rare combination of top-tier academic research (Ph.D., Impact Factor 8.9 lead author, patent inventor) and hands-on PyTorch implementation of Vision-Language and Vision-Language-Action foundation models (LoRA, Prompt Engineering, RAG) for complex real-world human behavior and retail/industrial action understanding.
- **2. Proprietary Occlusion-Proof Multi-Camera Tracking (CTA — 100% Accuracy):** Creator of the Customized Tracking Algorithm (CTA) that solves identity switching across crowded, multi-camera environments—proven both in 24/7 livestock farms (99.11% over 12-hour windows) and commercial security/retail systems (100% accuracy even with masks, eyeglasses, and sunglasses).
- **3. Full-Lifecycle Ownership — From Mathematical Formulation to Edge AI & Client Delivery:** End-to-end execution across algorithm design (e.g., ~25m zero-annotation calibrated keypoint fall detection, packing action recognition), hardware acceleration on NVIDIA Jetson (CUDA/TensorRT/ONNX), GUI/API development, and pre-sales customer installation.